

Lecture 4:

Numerical Differentiation and Integration



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University of California, Berkeley

**Numerical Methods for
Computational Science**

MSSE 273, 3 Units

Course Map

Week 1:	Introduction to Scientific Computing and Python Libraries
Week 2:	Linear Algebra Fundamentals
Week 3:	Vector Calculus
Week 4:	Numerical Differentiation and Integration
Week 5:	Solving Nonlinear Equations
Week 6:	Probability Theory Basics
Week 7:	Random Variables and Distributions
Week 8:	Statistics for Data Science
Week 9:	Eigenvalues and Eigenvectors
Week 10:	Simulation and Monte Carlo Method
Week 11:	Data Fitting and Regression
Week 12:	Optimization Techniques
Week 13:	Machine Learning Fundamentals



Outline

- Finite Differences
- Numerical Integration



Outline

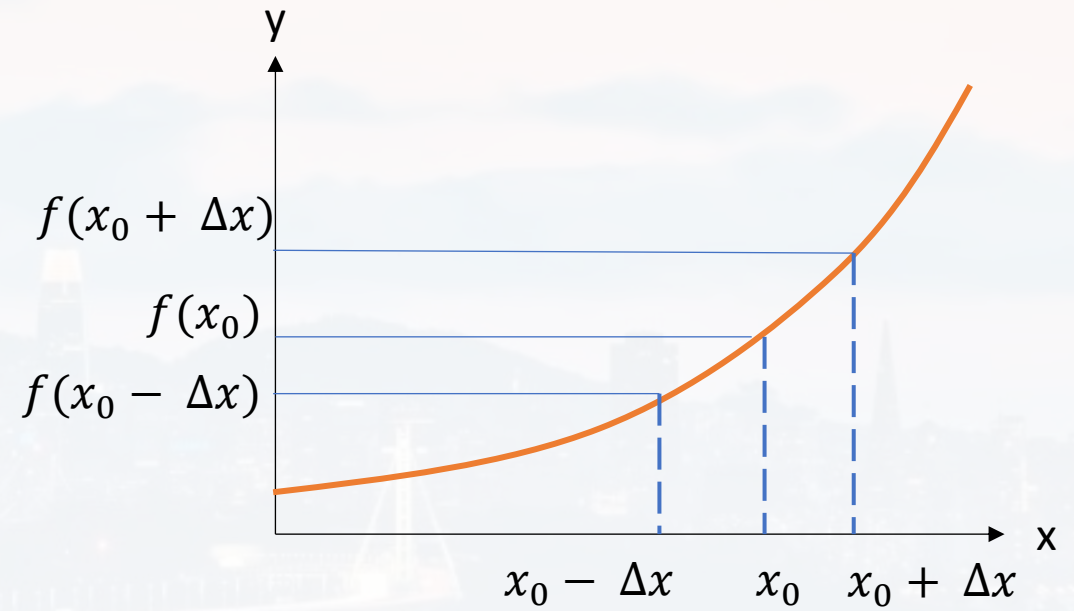
- Finite Differences

- Numerical Integration

slope of a function at $x = x_0$

$$\left. \frac{df^+}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

$$\left. \frac{df^-}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0) - f(x_0 - \Delta x)}{\Delta x}$$



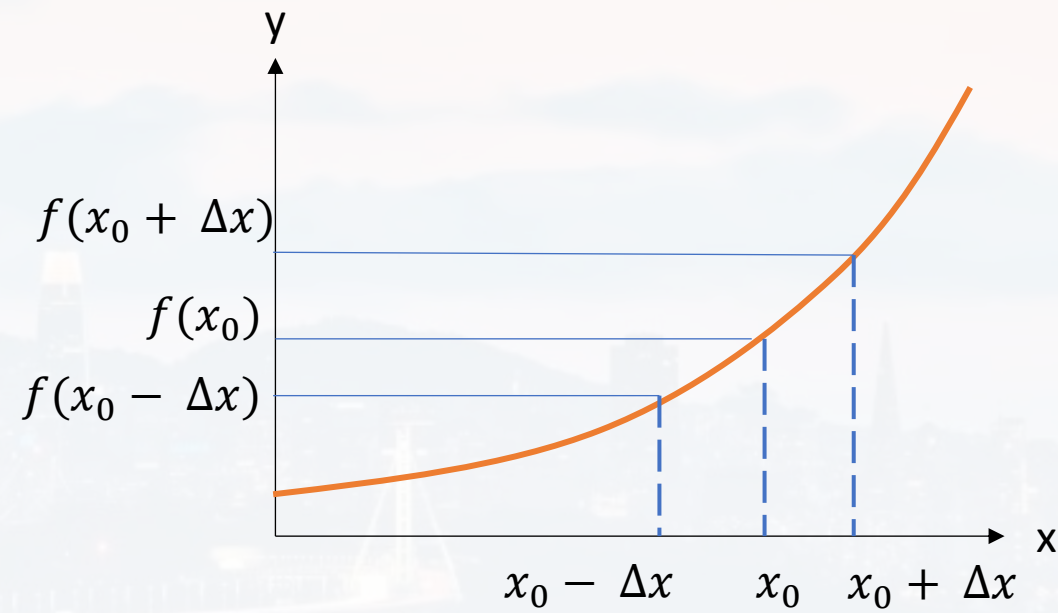
$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{2} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} + \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$

slope of a function at $x = x_0$

$$\left. \frac{df^+}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

$$\left. \frac{df^-}{dx} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0) - f(x_0 - \Delta x)}{\Delta x}$$



Δx

$$\Delta f = f(x_0 + \Delta x) - f(x_0)$$

finite differences

$$\Delta x$$

finite differences

$$\Delta f = f(x_0 + \Delta x) - f(x_0)$$

$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{2} \left(\left. \frac{df^+}{dx} \right|_{x=x_0} + \left. \frac{df^-}{dx} \right|_{x=x_0} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}$$

1st derivative at $x = x_0$

$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - 2f(x_0) + f(x_0 - \Delta x)}{\Delta x^2}$$

2nd derivative at $x = x_0$

often, we need to **model diffusion processes** (heat conduction, diffusion reaction, electromagnetism etc)

$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t)$$

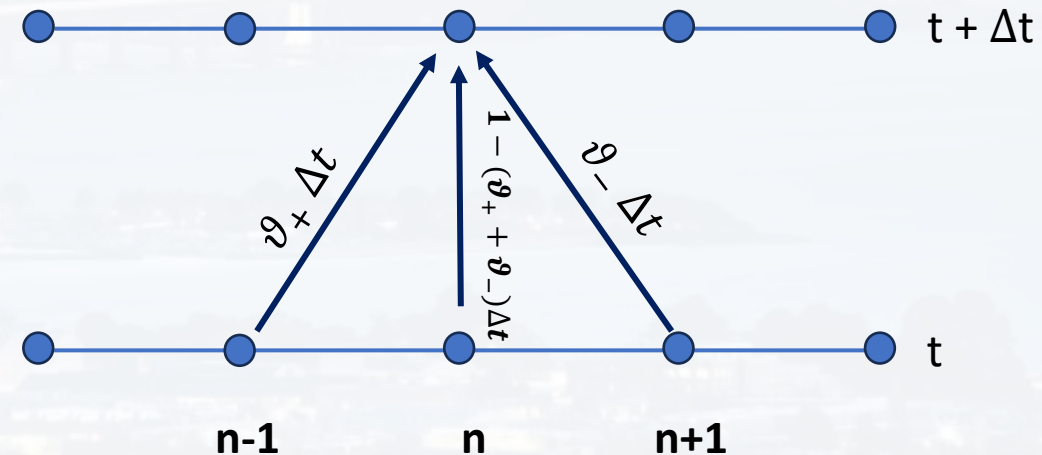
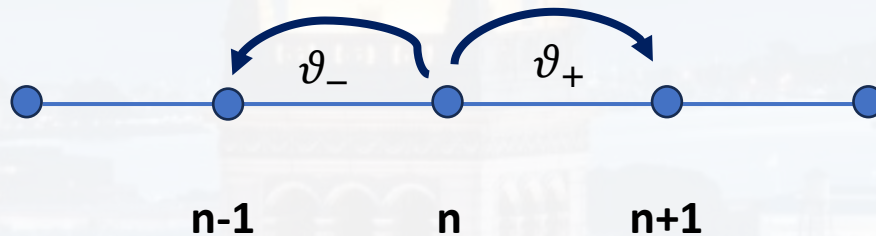
often, we need to **model diffusion processes** (heat conduction, diffusion reaction, electromagnetism etc)

$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t)$$

Fick's 2nd law

c : concentration
 D : diffusion constant
 $\Delta \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$: Laplace operator

n : state (e.g. location)
 ϑ_-, ϑ_+ : hopping rate (probability per time)

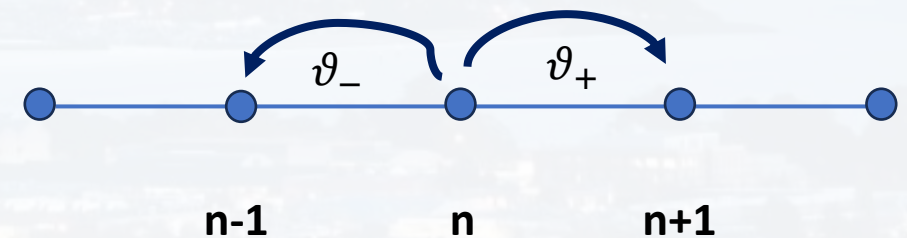


$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t) \quad \text{Fick's 2nd law}$$

The probability **$P(n, t)$** that the system is in state **n** at time **t**

$$\begin{aligned} \frac{\partial}{\partial t} P(n, t) = & P(\text{jump up} | \text{at } n-1) P(\text{at } n-1) \\ & + P(\text{jump down} | \text{at } n+1) P(\text{at } n+1) \\ & - P(\text{jump down} | \text{at } n) P(\text{at } n) \\ & - P(\text{jump up} | \text{at } n) P(\text{at } n) \end{aligned}$$

c :	concentration
D :	diffusion constant
$\Delta \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$:	Laplace operator
n :	state (e.g. location)
ϑ_-, ϑ_+ :	hopping rate (probability per time)

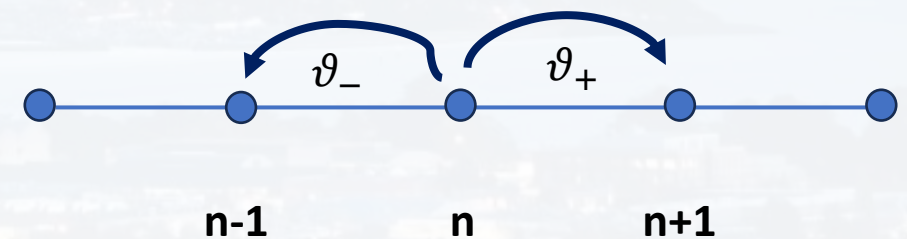


$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t) \quad \text{Fick's 2nd law}$$

The probability **$P(n, t)$** that the system is in state **n** at time **t**

$$\begin{aligned} \frac{\partial}{\partial t} P(n, t) = & \vartheta_+ P(n-1, t) \\ & + \vartheta_- P(n+1, t) \\ & - \vartheta_- P(n, t) \\ & - \vartheta_+ P(n, t) \end{aligned}$$

c :	concentration
D :	diffusion constant
$\Delta \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$:	Laplace operator
n :	state (e.g. location)
ϑ_-, ϑ_+ :	hopping rate (probability per time)



$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t) \quad \text{Fick's 2nd law}$$

The probability **$P(n, t)$** that the system is in state **n** at time **t**

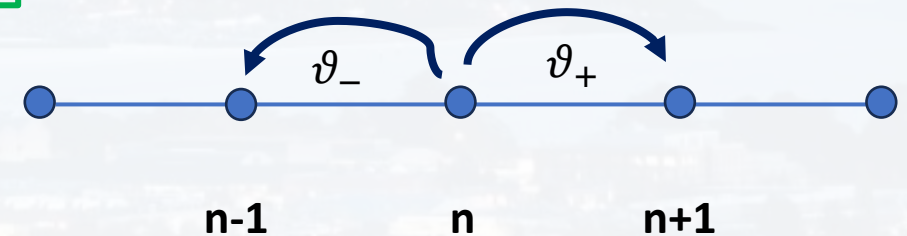
$$\frac{\partial}{\partial t} P(n, t) = \vartheta_+ P(n-1, t) + \vartheta_- P(n+1, t) - [\vartheta_- + \vartheta_+] P(n, t)$$

in case $\vartheta_- = \vartheta_+ = \vartheta$

$$\frac{\partial}{\partial t} P(n, t) = \vartheta P(n-1, t) + \vartheta P(n+1, t) - 2\vartheta P(n, t)$$

Master Equation

c :	concentration
D :	diffusion constant
$\Delta \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$:	Laplace operator
n :	state (e.g. location)
ϑ_-, ϑ_+ :	hopping rate (probability per time)



$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t)$$

$$\frac{\partial}{\partial t} P(n, t) = \vartheta [P(n-1, t) + P(n+1, t) - 2P(n, t)]$$

$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - 2f(x_0) + f(x_0 - \Delta x)}{\Delta x^2}$$

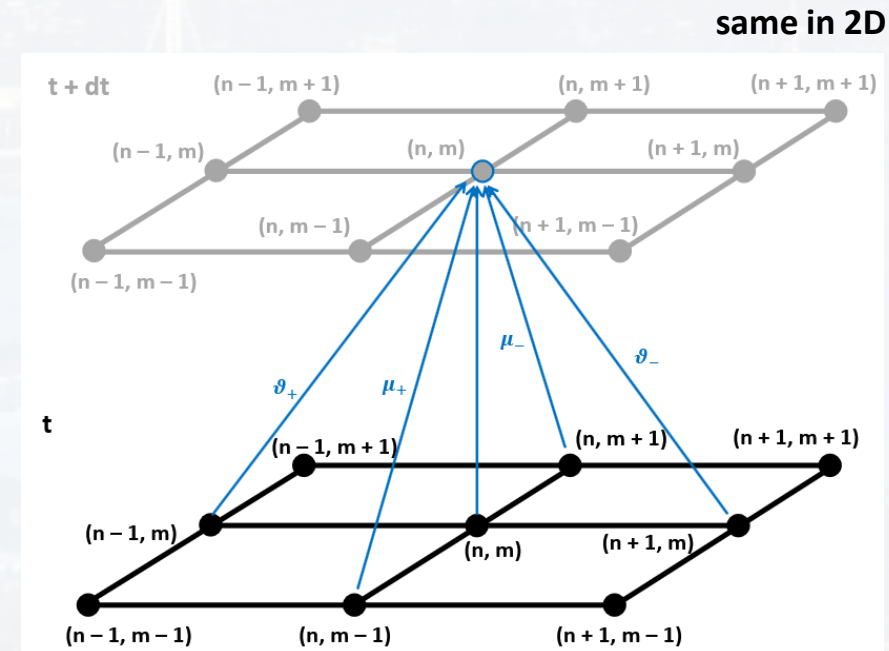
$$\frac{\partial}{\partial t} P(n, t) = \vartheta \frac{\partial^2}{\partial n^2} P(n, t)$$

$$\frac{\partial}{\partial t} P(n, t) = \vartheta \Delta P(n, t)$$

Fick's 2nd law

Master Equation

2nd derivative at $x = x_0$



$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t) \quad \text{Fick's 2nd law}$$

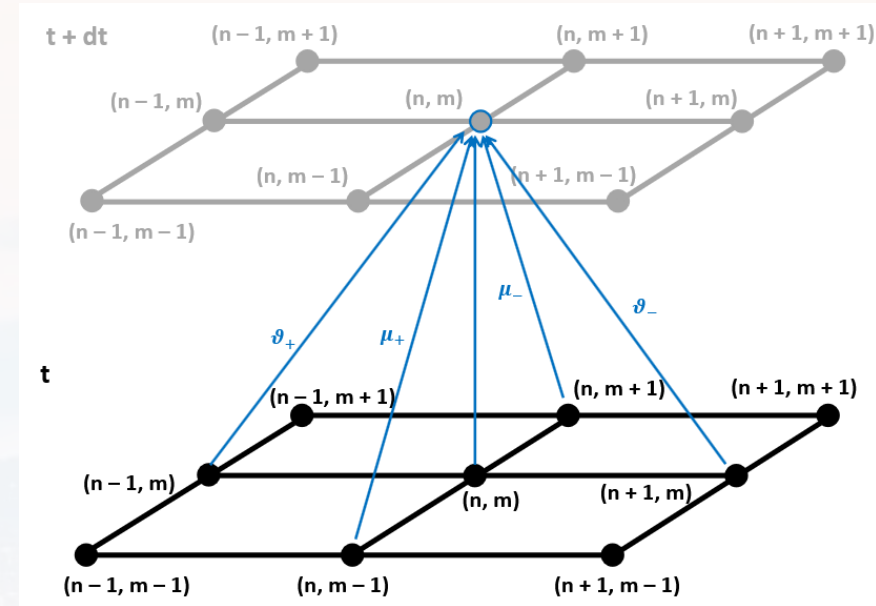
The Laplace operator indeed describes diffusion!

numerically:

$$\frac{c(x_0, y_0, t_0 + \Delta t) - c(x_0, y_0, t_0 - \Delta t)}{2\Delta t} = D \left[\right.$$

$$\frac{c(x_0 + \Delta x, y_0, t_0) - 2c(x_0, y_0, t_0) + c(x_0 - \Delta x, y_0, t_0)}{\Delta x^2} +$$

$$\left. \frac{c(x_0, y_0 + \Delta y, t_0) - 2c(x_0, y_0, t_0) + c(x_0, y_0 - \Delta y, t_0)}{\Delta y^2} \right]$$



$$\frac{\partial c(x, y, z, t)}{\partial t} = D \Delta c(x, y, z, t) \quad \text{Fick's 2nd law}$$

The Laplace operator indeed describes diffusion!

numerically:

$$c(x_0, y_0, t_0 + \Delta t) =$$

We can calculate c
in the **future**

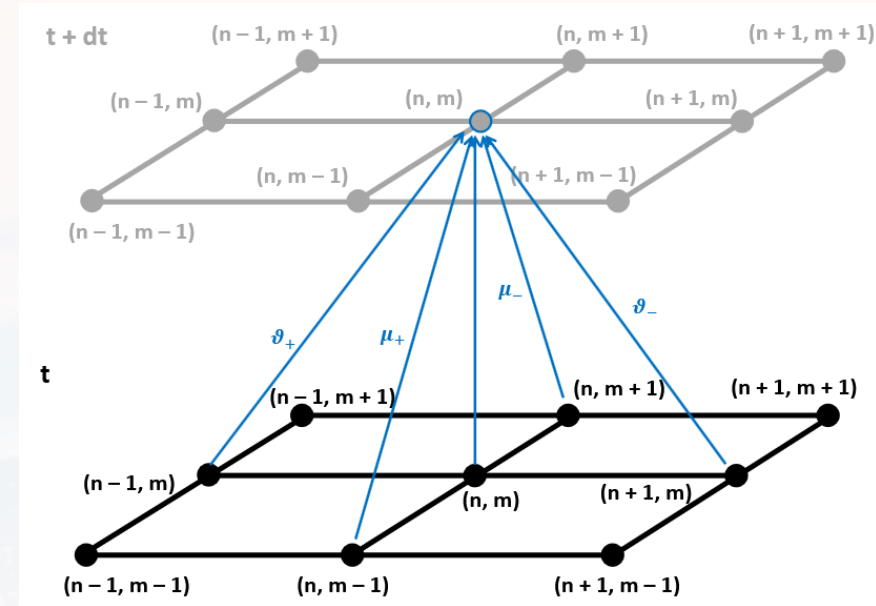
$$2\Delta t D \left[\frac{c(x_0 + \Delta x, y_0, t_0) - 2c(x_0, y_0, t_0) + c(x_0 - \Delta x, y_0, t_0)}{\Delta x^2} + \right.$$

$$\left. \frac{c(x_0, y_0 + \Delta y, t_0) - 2c(x_0, y_0, t_0) + c(x_0, y_0 + \Delta y, t_0)}{\Delta y^2} \right]$$

$$+ c(x_0, y_0, t_0 - \Delta t)$$

by using all adjacent
current values

and the immediate
past value

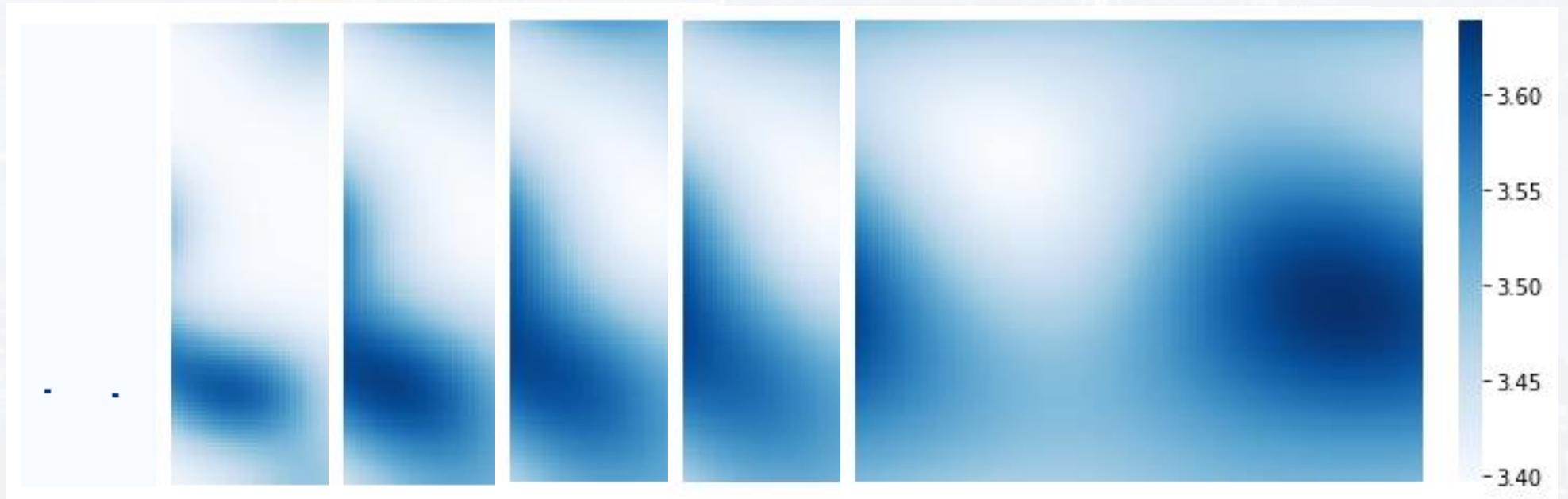
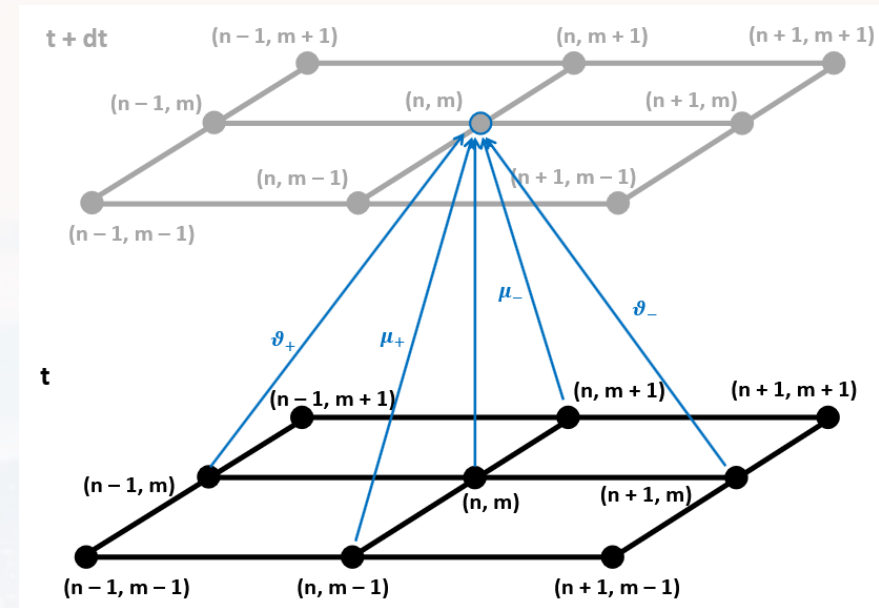


run the script Diffusion2D.py

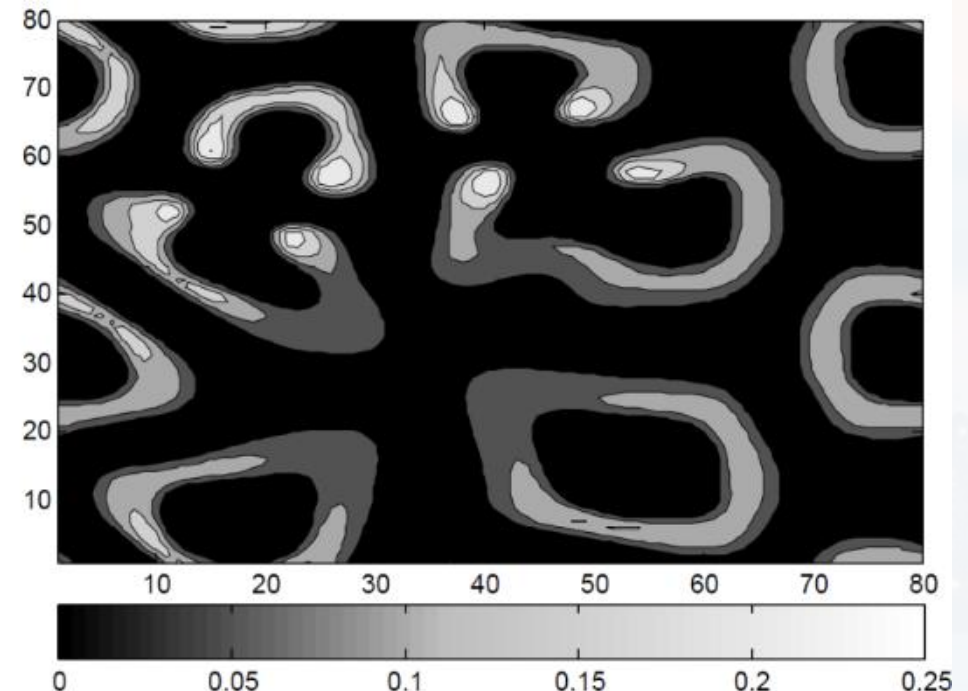
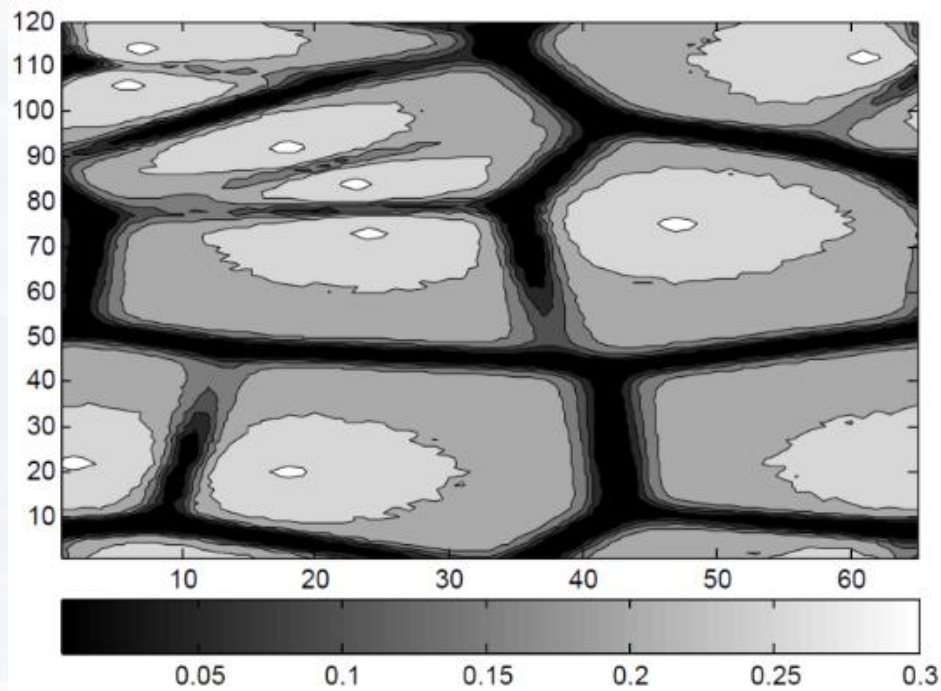
```
from Diffusion2D import *
```

```
D = Diffusion2D()
D.RunSimulation()
```

initial condition



Project I (Module 9): modelling fur and skin pattern:





Outline

- Finite Differences

- **Numerical Integration**



$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

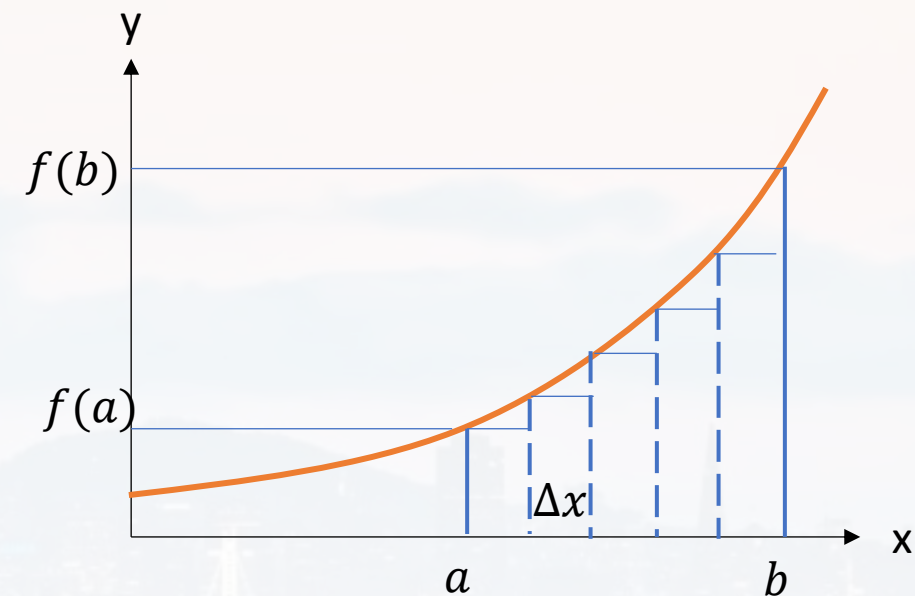
$$N = \frac{b - a}{\Delta x}$$

more accurate:

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i + 1) \Delta x)] \frac{\Delta x}{2}$$

error (for large N):

$$\varepsilon = -\frac{(b - a)^2}{12 N^2} [f'(b) - f'(a)] + O(N^{-3})$$



trapezoidal rule



$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} f(a + i \Delta x) \Delta x$$

$$N = \frac{b - a}{\Delta x}$$

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i + 1) \Delta x)] \frac{\Delta x}{2}$$

trapezoidal rule

even more accurate:

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_{i=0}^{N-1} [f(a + i \Delta x) + f(a + (i + 1) \Delta x) + 4 f(a + i \Delta x/2)] \frac{\Delta x}{6}$$

Simpson rule

Note: there are different Simpson rules, depending on how many subintervals are included

Newton-Cotes Equations

approximation

error

$$\frac{1}{2} \Delta x (f_i + f_{i+1})$$

$$\varepsilon \sim \frac{\Delta x^3}{12}$$

trapezoidal

$$\frac{1}{6} \Delta x (f_i + f_{i+2} + 4f_{i+1})$$

$$\varepsilon \sim \frac{\Delta x^5}{90}$$

Simpson

$$\frac{1}{8} \Delta x (f_i + 3f_{i+1} + 3f_{i+2} + f_{i+3})$$

$$\varepsilon \sim \frac{3 \Delta x^5}{80}$$

Simpson 3/8

$$\frac{1}{90} \Delta x (7f_i + 32f_{i+1} + 12f_{i+2} + 32f_{i+3} + 7f_{i+4})$$

$$\varepsilon \sim \frac{8 \Delta x^7}{945}$$

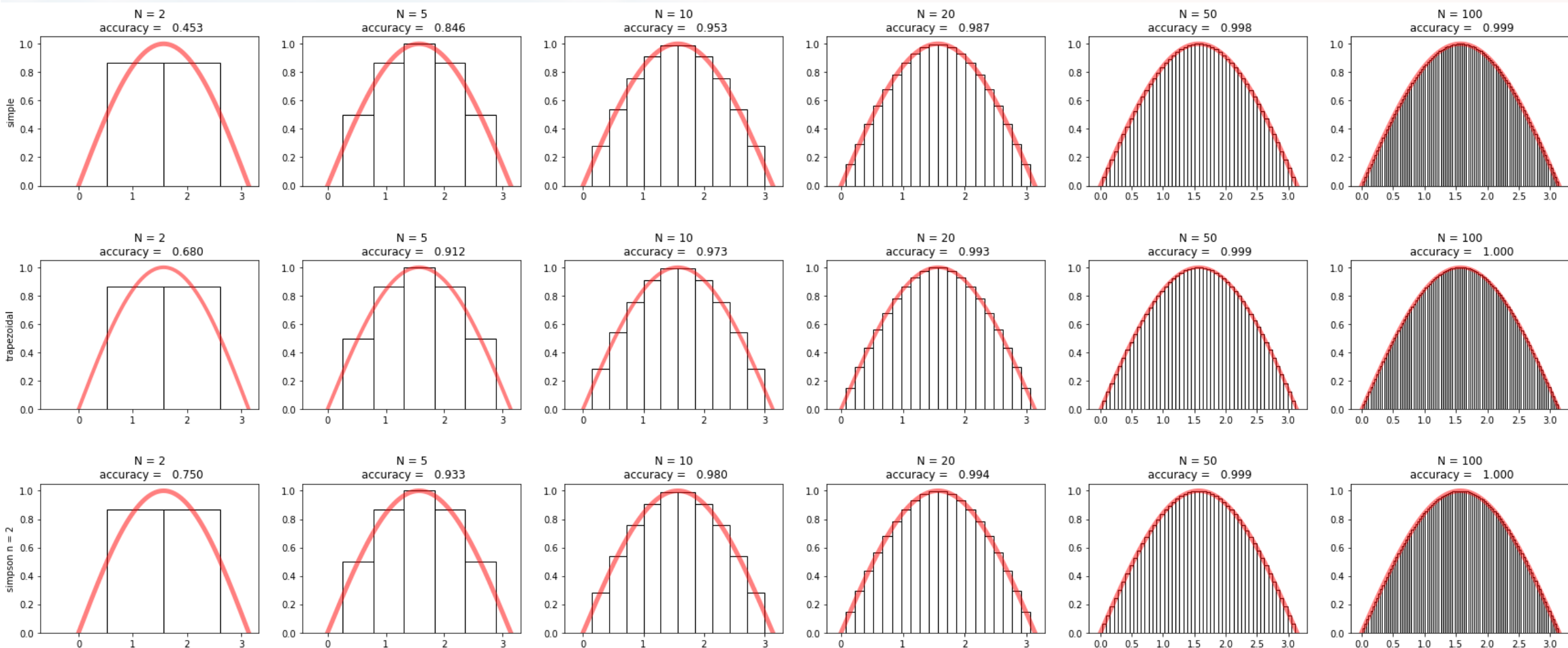
Boole

Note: i here refers to subinterval **within** Δx



run the function `IntegrationAccuracy.py`

integrating $\sin(x)$



SciPy

```
from scipy.integrate import *
```

```
simpson  
trapezoid  
quad
```

...and more.
See lecture exercise!

```
I = simpson(y, x)
```




Thank you very much for your attention!

